

Question number	Answer	Additional Guidance	Mark
	There is no (significant) difference between the diameter of control stems and experimental stems	Accept no difference between the diameters of the two groups	(1)

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	- suitable table format with data (1) - correct column headings with units (1) - means correctly calculated (1)	Example table <table><tr><th colspan="2">Stem diameter / mm</th></tr><tr><th>Control stems</th><th>Bent stems</th></tr><tr><td>17.15</td><td>17.34</td></tr><tr><td>17.52</td><td>18.16</td></tr><tr><td>17.57</td><td>17.61</td></tr><tr><td>17.03</td><td>18.14</td></tr><tr><td>17.26</td><td>17.64</td></tr><tr><td>17.31</td><td>18.14</td></tr><tr><td>17.09</td><td>17.95</td></tr><tr><td>17.38</td><td>17.78</td></tr><tr><td>17.41</td><td>17.69</td></tr><tr><td>17.21</td><td>17.84</td></tr><tr><td>17.34</td><td>17.99</td></tr><tr><td>17.16</td><td>17.95</td></tr><tr><td>17.15</td><td>17.89</td></tr><tr><td>17.32</td><td>18.16</td></tr><tr><td>17.60</td><td>18.22</td></tr><tr><td>17.3(0)</td><td>17.9(0)</td></tr></table> Not 17 18 as mean values Means can be to two places or one but not mixed / Means do not have to be in the table Ignore errors in table data eg 17.6 instead of 17.60 Not MP2 if units in the body of the table	Stem diameter / mm		Control stems	Bent stems	17.15	17.34	17.52	18.16	17.57	17.61	17.03	18.14	17.26	17.64	17.31	18.14	17.09	17.95	17.38	17.78	17.41	17.69	17.21	17.84	17.34	17.99	17.16	17.95	17.15	17.89	17.32	18.16	17.60	18.22	17.3(0)	17.9(0)	
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3c	- bar graph with labels, units and linear scale (1) - means plotted correctly (1) - range bars plotted correctly (1)	mean diameter /mm and group / eq BAR GRAPH MUST START AT ZERO AND NO BROKEN SCALE Allow ECF from 3bi 17.34 -18.22 17.03 - 17.60 MP2 and 3 can be awarded for line graph	(3)

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3di	- correct substitution of given $(SA)^2$ and $(SB)^2$ (1) - correct answer (1)	Allow wrong means used for 2 marks ie 0.03 and 0.06 7.75 Accept 7.746 / 7.7459 MP2 only for 34.6 / 34.64 / 34.641	(2)

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3d ii	- the calculated value of t (7.75) is more than the critical value 2.05 (at p=0.05) (1) - therefore reject the {null hypothesis/H_0}, there is a difference between the diameter of control and experimental stems (1)	Allow use of any quoted value as the calculated value (from 3d i so reverse statements) Allow use of p=0.01 and critical value of 2.76 Allow reverse statement if calculated value is stated / shown as less than 2.05	(2)

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3e	An explanation that includes two of the following: - the {seeds/plants} may have been {different varieties / genetically different} (1) - plants may bend for other reasons eg phototropism / light direction (1) - the range bars overlap	Ignore IAA/auxins	(2)